

Detailed Methodology

1. Data Collection and Preprocessing

1.1 Dataset Acquisition

- **Source:** Collect a dataset of student-generated text (e.g., forum comments, discussion posts, feedback) from Vietnamese online education platforms or publicly available Vietnamese NLP datasets focused on educational/social media text (e.g., UIT-VSFC, if applicable for fine-grained emotion, or a curated corpus).
- **Annotation:** Ensure the dataset is annotated for fine-grained emotion categories (e.g., *excitement, anxiety, frustration, confusion, boredom, satisfaction*) and general sentiment (positive, negative, neutral).

1.2 Preprocessing Pipeline

- **Standardization:** Apply Vietnamese text normalization (e.g., converting common abbreviations, resolving inconsistent spellings, converting Unicode to canonical forms).
- **Tokenization:** Utilize the appropriate tokenizers corresponding to the baseline models (e.g., [PhoBERTTokenizer](#) for PhoBERT).
- **Cleaning:** Remove noise such as URLs, excessive special characters, and potentially PII (Personally Identifiable Information).

2. Baseline Model Training and Evaluation

2.1 Model Selection

Train and evaluate three state-of-the-art Vietnamese transformer models:

- **PhoBERT (Base/Large)**
- **VisoBERT**
- **CafeBERT**

2.2 Training Protocol

- **Task:** Multi-class fine-grained emotion classification.
- **Configuration:** Use standard configurations (e.g., max sequence length = 128, batch size = 16/32, learning rate schedule, AdamW optimizer).
- **Validation:** Employ 5-fold cross-validation or a fixed 80/10/10 split for train/validation/test sets.
- **Hyperparameter Tuning:** Conduct light tuning (e.g., learning rate, number of epochs) based on performance on the validation set.

2.3 Evaluation and Benchmarking

- **Metrics:** Calculate Macro F1, Micro F1, and Accuracy on the held-out test set.
- **Output:** Establish a baseline performance table for comparison with subsequent methods.

Model	Macro F1 (Baseline)	Micro F1 (Baseline)	Accuracy (Baseline)
PhoBERT			
VisoBERT			
CafeBERT			

3. LLM-Based Data Augmentation

3.1 Minority Class Identification

- Analyze the distribution of emotion labels in the training set to identify low-frequency (minority) categories that suffer from low baseline performance.

3.2 Prompt Engineering for Synthetic Data Generation

- **Objective:** Generate high-quality, linguistically authentic Vietnamese sentences belonging to the identified minority emotion categories.
- **LLM Tool:** Utilize GPT-4 or GPT-5 (via API) for generation.
- **Prompt Structure:** Develop controlled, structured prompts that guide the LLM:
 - **Context Specification:** "Generate a Vietnamese student comment in the context of an online lecture about X topic."
 - **Emotion Target:** "The comment must express the emotion: [Minority Emotion Label] (e.g., confusion, excitement)."
 - **Format Constraint:** "Ensure the language is natural and idiomatic Vietnamese."
 - **Intensity Control:** Include variations in prompts to generate content with varying emotional intensities.
- **Validation:** Implement a rigorous validation step:
 - **Automated Filtering:** Filter out duplicates or sentences too short/generic.
 - **Manual/Expert Review:** A small subset of generated data must be manually reviewed by a Vietnamese speaker/NLP expert to verify linguistic quality and label correctness, preventing catastrophic label leakage or bias.

3.3 Augmentation and Retraining

- **Augmentation Strategy:** Balance the dataset by adding synthetic data to minority classes until a predetermined threshold or balanced distribution is achieved.

- **Retraining:** Fine-tune the best-performing baseline transformer model (e.g., the best of PhoBERT/VisoBERT/CafeBERT) using the combined original and synthetic training data.

3.4 Evaluation of Augmentation

- **Metrics:** Re-evaluate the retrained model on the *original* test set.
- **Analysis:** Compare the Macro F1 (specifically focusing on gains in minority class F1 scores) and overall Accuracy against the baselines.

4.4.1 Component Selection

- **BERT Components:** Select the top-three performing transformer models after data augmentation (Model-Augmented 1, 2, and 3) to be the candidate predictors.
- **LLM Component (Meta-Learner):** Use LLMs to act as a meta-learner/selector.

4.3 Ensemble Strategy

- **Input:** Predictions (label and confidence score) from:
 - The top three Model-Augmented transformer models (BERT 1, BERT 2, BERT 3).
- **Selection Method (LLM Meta-Learning):** The LLM component is used as a meta-learner to select the final prediction.
- **Selection Protocol:**
 - **Confidence-Based Selection:** The LLM will be provided with the input text and the predictions (label and confidence score) from the three BERT models. The LLM's task is to analyze the input and the associated confidence scores, and select the prediction from the BERT model with the highest confidence score, thereby acting as a filter or arbiter.
- **Final Evaluation:** Evaluate the selection model on the test set.

Selection Type	Macro F1	Micro F1	Accuracy
LLM-Based Confidence Selection			
PhoBERT			
PhoBERT-Augmented			

CafeBERT			
CafeBERT-Augmented			
VisoBERT			
VisoBERT-Augmented			

5. Topic Extraction (Supporting Task)

5.1 Traditional Baseline

- **Method:** Utilize **BERTopic** on the student comment data.
- **Configuration:** Use a suitable Vietnamese embedding model (e.g., PhoBERT embeddings) and default BERTopic settings (UMAP for dimensionality reduction, HDBSCAN for clustering).
- **Evaluation:** Calculate the **Coherence Score (Cv)** for the extracted topics.

5.2 LLM-Assisted Topic Generation

- **Tool:** Implement an LLM pipeline using **LangChain** or direct API calls.
- **Process (Map-Reduce Style):**
 - Map:** Divide the entire document corpus into manageable chunks. Use GPT-4/GPT-5 to generate 3-5 core topics for each chunk.
 - Reduce/Refine:** Consolidate the lists of topics generated in the Map step using a second LLM prompt ("Group these lists of topics into 10 cohesive, high-level themes, ensuring minimal overlap and clarity.").
 - Refinement:** Ask the LLM to provide a descriptive name and a brief summary for each final topic.

5.3 Topic Evaluation

- **Metrics:**
 - **Coherence Score (Cv):** Calculate the Cv score for both BERTopic and LLM-assisted topics for direct comparison.
 - **Human Interpretability Score:** Recruit 3-5 human evaluators (e.g., academics or educators) to rate the top 10 topics from each method on a 5-point Likert scale based on clarity, relevance to the educational context, and distinctness.

Method	Coherence Score (Cv)	Human Interpretability Score (Avg)
BERTopic		
LLM-Assisted Topic Generation		

6. Combined Impact Analysis

- **Synthesis:** Analyze the overall performance gain from the sequential application of methodologies:
 - Baseline / Augmentation / Ensemble.
- **Robustness Analysis:** Assess the final ensemble model's performance on a per-class basis to determine if the combined approach effectively addressed the vulnerability of minority or ambiguous emotion categories.
- **Integration:** Demonstrate how the extracted topics (LLM-assisted) can provide contextual support by visualizing the distribution of high-anxiety or high-confusion comments across specific themes (e.g., "Technical Issues," "Exam Preparation," "Lecture Content Quality"). This addresses the goal of providing "supporting context."